

Deloitte.



Design Challenge

UTS Engineering Society

Engineering



*WE ACKNOWLEDGE THE TRADITIONAL OWNERS AND
CUSTODIANS OF THE LAND ON WHICH WE MEET
TODAY AND PAY OUR RESPECT TO ELDERS PAST,
PRESENT AND EMERGING.*

Agenda

- 01 Introduction
- 02 Design Challenge: KapD
- 03 Timelines and Evaluations
- 04 Q&A
- 05 Next Steps



01 Introduction



Our Team



Engineering



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Who is Deloitte?

Deloitte is a large company that helps **businesses** and **governments** solve tough problems. We give **advice** and **support** in areas like **finance, technology, strategy, risk**, and more.

Engineering, AI & Data

We are an integrated team of industry specialists, technologists, and operations transformation experts within Deloitte’s Technology & Transformation business unit. We work with our clients to transform and run their **technology platforms**, optimise their **operations** and help them take advantage of **new technologies**, drive **innovation**, improve **financial performance** and accelerate **speed to market**.

Engineering, AI & Data Market Offerings

ARTIFICIAL INTELLIGENCE & DATA	ENGINEERING	ENGINEERING AS A SERVICE	OPS, INDUSTRY & DOMAIN SOLUTIONS
Transformation & Data Services	Cloud & Engineering Strategy & Architecture	Managed Analytics & AI/ML Ops	Operations Excellence
AI & Data Platforms	Platform Development & Integration	Engineering Foundry & Managed Services	Banking Solutions
AI & Analytical Solutions	Cloud & Infrastructure Engineering	Cloud Managed Services	Insurance Solutions
Generative AI	Secure Network & Operations Engineering	Platform Managed Services	GPS Solutions
AI Strategy	Application Modernisation	Data Ops	Healthcare Solutions
Intelligent Automation	Quality Engineering		Process Bionics & Digital Solutions

Introduction to the Hackathon

This hackathon has been designed to provide students with the opportunity to think like a consultant and turn their ideas into solutions that matter.



What's the purpose?

- Give students a chance to tackle a realistic, multidisciplinary business problem that brings together **engineering, data, strategy, and operations thinking**
- Designed to help students experience how complex problems are approached in practice — by balancing **technical design, data reliability, user needs, and business value** in one coherent solution.



Why is it relevant?

- Designed to mirror how consulting teams work on client problems: understanding the context, identifying priorities, and developing a solution that is both **strategically compelling and operationally feasible**.
- Students are not only being asked what the solution is, but also **who it is for, how it works, how it will be governed, and how it creates business and clinical value**
- The deliverable mirrors **real consulting work**: turning ambiguity into a clear recommendation



What's in it for me?

- Students will build experience in **structured problem-solving**, working in teams to shape a solution and communicate it clearly through a professional presentation pack
- Develop practical skills across areas such as **product thinking, solution design, data governance, and operational feasibility**, which are all core parts of the challenge brief
- Strengthen ability to **think critically, collaborate** under time constraints, and present **recommendations** in a way that is **clear, persuasive**, and audience-focused.

02 Design Challenge

KapD

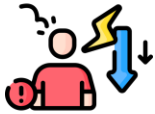


Problem Context

Kappa Diagnostics (KapD) is a global medical devices company producing diagnostic instruments and informatics solutions used by hospitals, blood banks, and laboratories. KapD's products include physical diagnostic machines and software platforms that manage and distribute test results across hospital systems.

One example is the Sigma immunoassay unit, which analyses blood samples, and its integration with SigmaD, a diagnostic hub (LIMS) that manages test orders and results and shares them with clinical teams in near real time via standard healthcare messaging.

Healthcare providers are under increasing pressure to:



- Deliver faster diagnostic turnaround times
- Reduce operational inefficiencies
- Improve interoperability across systems
- Support growing diagnostic workloads

Many hospitals currently operate disconnected systems where:



- Diagnostic devices are siloed
- Laboratory systems vary between hospitals
- Data sharing is inconsistent
- Downtime can disrupt clinical operations

Kappa Diagnostics (KapD) wants to reposition itself from a device manufacturer into a provider of integrated digital healthcare platforms.

KapD has engaged your consulting team to design a next-generation connected diagnostics solution.

Your proposal will be reviewed by:

- Product executives
- Engineering leaders
- Clinical operations stakeholders
- Hospital procurement representatives

Your Task

You will work in teams of 3 – 4 to design a presentation pack, demonstrating a packaged KapD offering that combines:

- Existing diagnostic devices
- Laboratory software
- A new Digital Services Layer

Your nominated group leader must upload the presentation to the allocated **teams channel** by **Sunday, 23rd August, 11:59pm.**

Note, we are not looking for a specific number of slides – but rather a sensible number to allow for great storytelling.

The solution should improve **connectivity**, **reliability**, **clinical usability**, **operational visibility** and **diagnostic data trust**.



Product Strategy

01

Explain:

- Target customers
- Problems being solved
- Why the bundled solution creates business and clinical value
- Competitive differentiation

Possible customer groups:

- Large hospital networks
- Pathology laboratories
- Regional healthcare providers
- Blood banks
- Emergency care providers

System Architecture

02

Design a high-level architecture showing:

- Diagnostic devices
- Data ingestion
- Integration layers
- Hospital systems
- User interfaces
- Analytics/reporting

Consider:

- Real-time vs batch data flows
- Cloud vs on-premise systems
- Multi-site deployments
- Scalability
- Testing

Data Governance

03

Address:

- Data quality
- Message reliability
- Duplicate handling
- Auditability
- Access controls
- Patient safety considerations

Operational Feasibility

04

Show how the platform supports:

- Laboratory technicians
- Clinicians
- Hospital administrators
- IT operations teams

Include:

- Dashboards
- Alerts
- Operational workflows
- Reporting capabilities

03 Timelines and Evaluations



Marking Criteria

Criterion	Poor (1)	Basic (2)	Good (3)	Excellent (4)
Presentation & Overall Solution	Unclear presentation, significant missing elements, solution does not address core requirements or demonstrates little understanding of improving connectivity, reliability, etc..	Mostly clear and concise, covering most required presentation elements, good overall solution with minor areas for improvement in combining KapD offerings or clearly articulating the benefits.	Clear and concise presentation including problem framing, product strategy, architecture, data flow, risks/trade-offs, recommendations, and a roadmap. The proposed KapD offering is coherent and largely complete, and effectively demonstrates improvement in most areas like connectivity, reliability, etc.	Clear and concise presentation including problem framing, product strategy, architecture, data flow, risks/trade-offs, recommendations, and a roadmap. The proposed KapD offering is highly coherent and complete, clearly demonstrating significant improvements in connectivity, reliability, clinical usability, operational visibility, and diagnostic data trust.
Product Strategy	Fails to identify target customers, problems solved or provide any clear value or differentiation.	Superficial identification of customers/problems, weak value proposition or differentiation.	Most elements identified and explained well, with some minor gaps in justification for target customers, problems solved, business/clinical value, and competitive differentiation.	Clearly identifies target customers and the problems being solved. Provides a compelling explanation of how the bundled solution creates business and clinical value and articulates strong competitive differentiation.
System Architecture (Engineering)	Incomplete architecture, no consideration of key factors (e.g., real-time vs batch, cloud vs on-premise), missing or unhelpful diagrams.	Architecture misses some components or considerations. Diagrams are basic or unclear.	Architecture generally covers components and considerations well. Diagrams are mostly clear.	High-level architecture showing diagnostic devices, data ingestion, integration layers, hospital systems, user interfaces, and analytics/reporting. Strongly considers real-time vs batch data flows, cloud vs on-premise systems, multi-site deployments, and scalability. Includes clear architecture diagrams and sequence diagrams.
Data Reliability & Governance (AI&D)	Fails to address most aspects of data quality, message reliability, duplicate handling, auditability, access controls, or patient safety.	Superficial addressing of some aspects of reliability and governance or misses several key areas.	Addresses most aspects of data quality, message reliability, duplicate handling, auditability, access controls, and patient safety considerations with good detail.	Comprehensively addresses data quality, message reliability, duplicate handling, auditability, access controls, and patient safety considerations.
Operational Feasibility	Little to no demonstration of support for user groups, missing key features (dashboards, alerts, workflows, reporting) and diagrams.	Limited support for user groups, few features included, diagrams are basic or unclear.	Shows good support for laboratory technicians, clinicians, hospital administrators, and IT operations teams and includes most features (dashboards, alerts, operational workflows, reporting capabilities), with mostly clear diagrams.	Clearly shows how the platform supports laboratory technicians, clinicians, hospital administrators, and IT operations teams. Includes relevant dashboards, alerts, operational workflows, and reporting capabilities, along with clear operational workflows and user journeys diagrams.

Timeline



Hackathon Kick-off

Kick-off session where case study details, materials and instructions are shared.



Drop-in Session

Set up virtually, this is an opportunity for students to ask questions and receive clarification/feedback.



Submission Date

Final presentations must be submitted by EOD as per the submission guidelines.



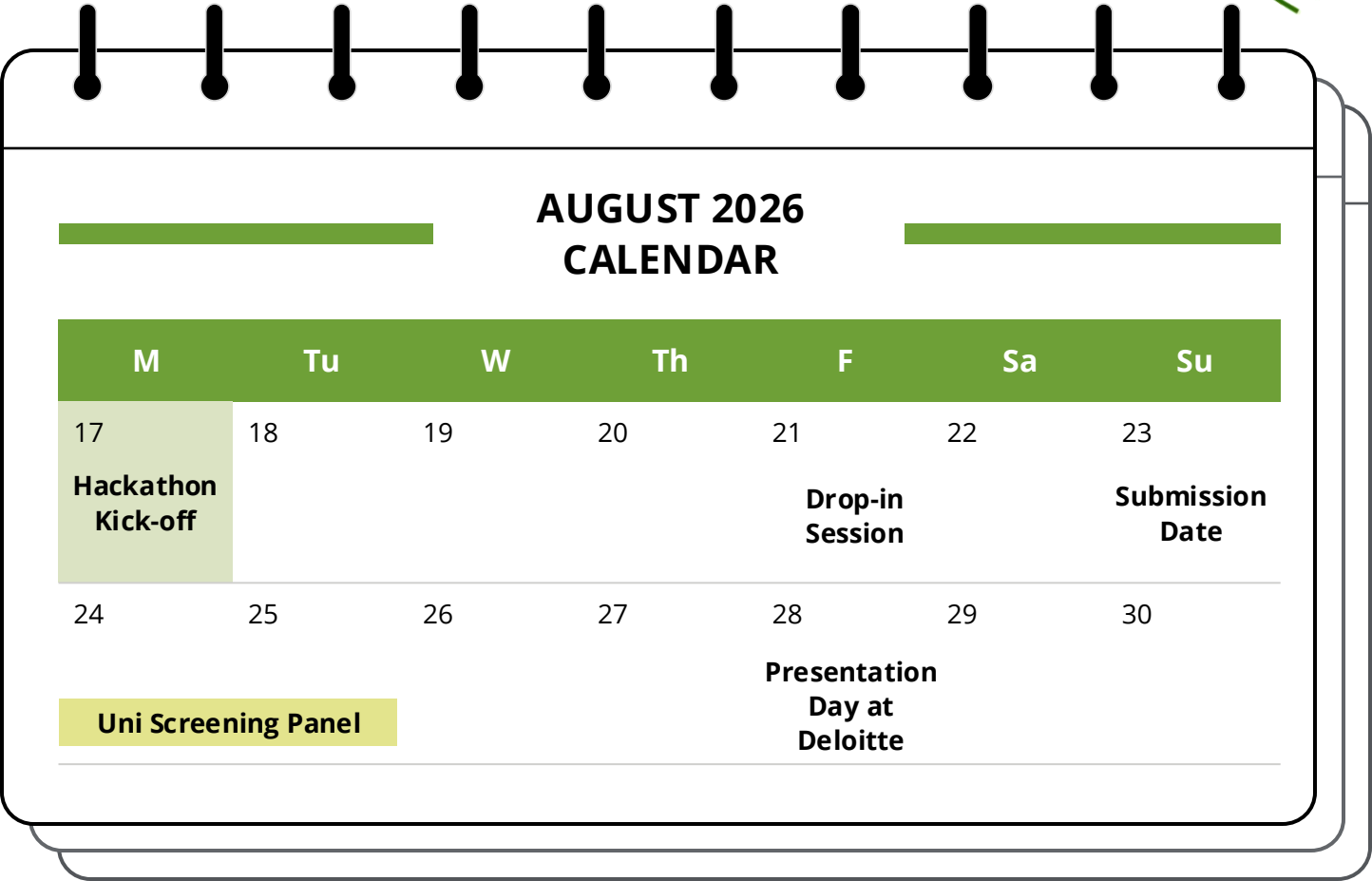
Uni Soc Screening Panel

Uni Society leads will screen all submissions and select the **top 5 groups** to progress to final round.



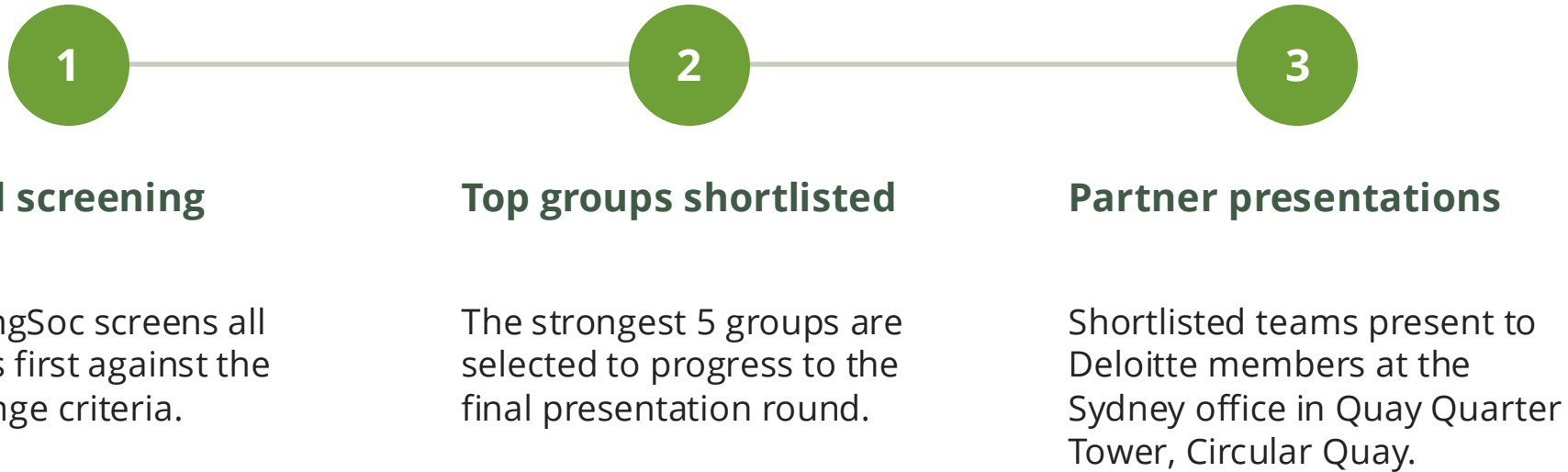
Presentation Day at Deloitte

Selected groups will present to a judging panel at the Deloitte office.



Final Presentation Evaluation

Entries are screened first by UTS Engineering Society (UTS EngSoc) before shortlisted teams present in person at Deloitte's Sydney office.

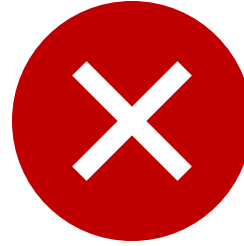


Use of AI

You may choose to use AI to support your submission but be sure to use it responsibly. Below are some good and bad uses of AI:



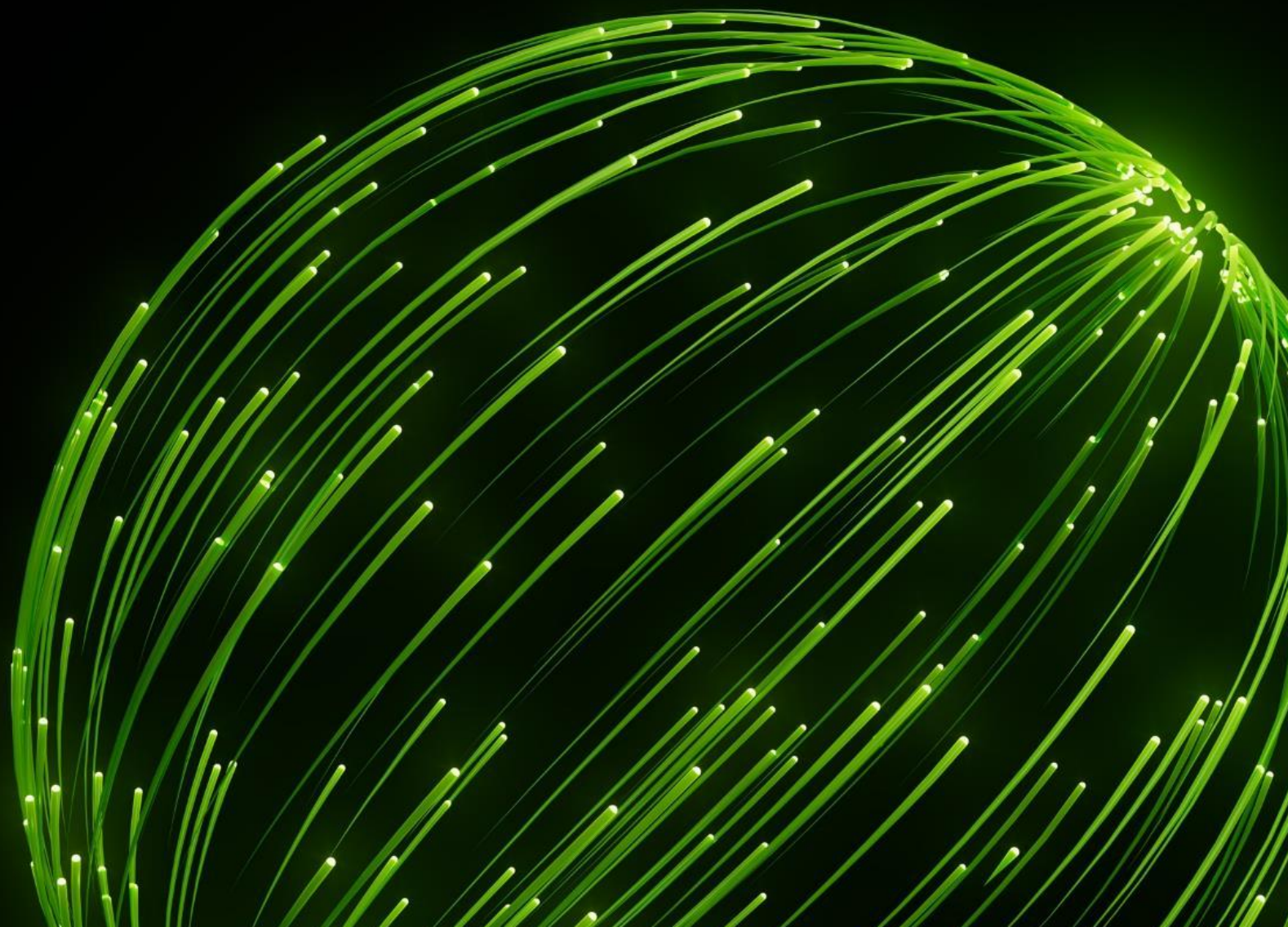
- ✓ Do your own research into the problem context
- ✓ Ask AI to answer specific questions relating to the task
- ✓ Validate and understand the outputs before including in submission
- ✓ Only provide AI tools the data that is necessary for the task
- ✓ Use AI to create a submission that demonstrates **YOUR** understanding and thinking



- × Expect to win the Design Challenge with 100% generated AI output.
- × Put the whole Design Challenge into an LLM
- × Briefly look through the output to see if it “looks right”
- × Directly submit the output AI provides with minimal changes (or none)
- × Demonstrate the understanding and “thinking” of an LLM in your submission

You will be assessed by Senior Leaders working in the AI space.
They can tell the difference between legitimate work and AI slop.
Especially when they start asking you questions about your work.

Q & A



Next Steps



Next Steps

Form your teams and start planning!

Don't forget to prepare questions (if required) for the upcoming drop-in session.

AUGUST 2026 CALENDAR						
M	Tu	W	Th	F	Sa	Su
17 Hackathon Kick-off	18	19	20	21 Drop-in Session	22	23 Submission Date
24 Uni Screening Panel	25	26	27	28 Presentation Day at Deloitte	29	30